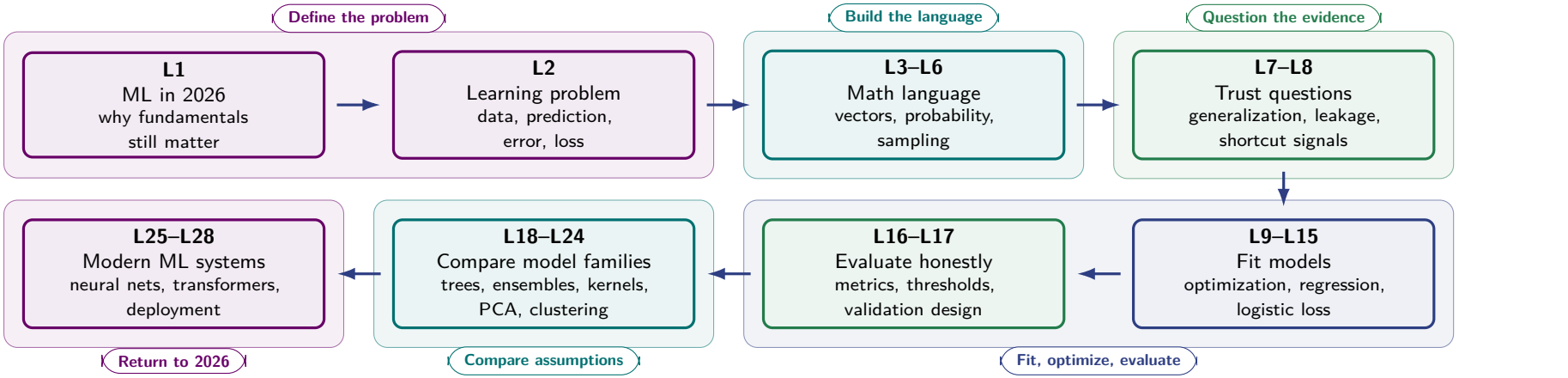
Course Thoughtmap

One semester-long journey from data to accountable ML



Habit 1: name the object
Before computing, ask whether the symbol is a scalar, vector, matrix, random variable, model, loss, metric, or decision.

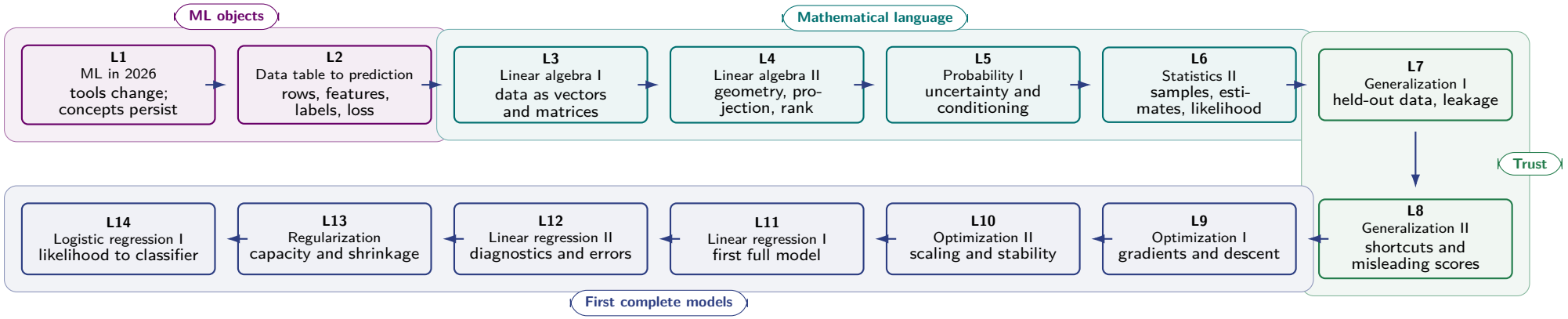
Habit 2: name the objective
Every algorithm is trying to reduce or improve something. State that quantity and the assumptions behind it.

Habit 3: name the failure mode
A high score can still be wrong because of leakage, shortcuts, distribution shift, weak baselines, or poor deployment context.

Semester spine: measurements become mathematical objects; objects define objectives; objectives lead to algorithms; algorithms demand evaluation; evaluation becomes responsibility when ML meets the real world.

Lecture Flow: Foundations To First Models

Lectures 1–14



After L1–L2, students know what the ML object is before heavy notation arrives.

L3–L6 build the language needed to make ML statements precise.

L7–L10 ask: when should we trust a computed answer?

L11–L14 turn notation into the first complete supervised models.

Lecture Flow: Model Families To ML Systems

Lectures 15–28

